Set of Exercises

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1 Introduction

1.1 Exercise 1

1.2 Exercise 2

see Code/Exercises/Exercise2.py for the Code

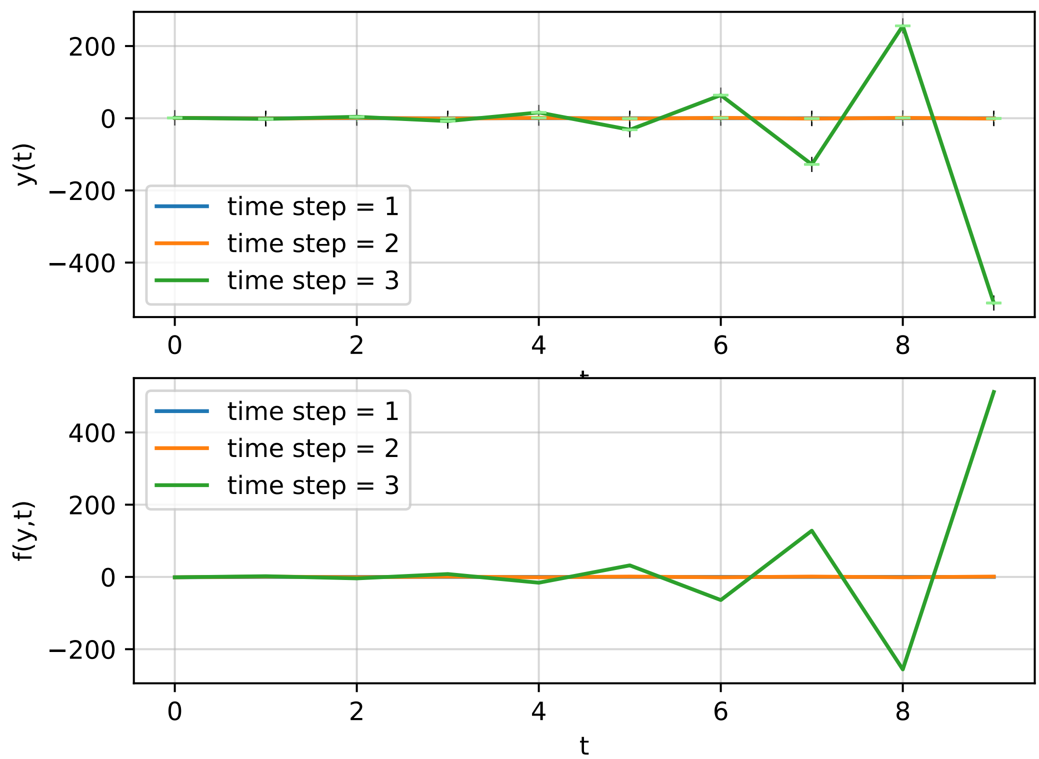

Figure 1: Figure showing the function $\frac{dy}{dt} = -y$ analysed numerically using the forward, or explicit Euler method with three different time steps. The upper pannel is the function and the lower pannel, its slope.

1.3 Exercise 3

Reusing the code from Exercise II, plus the code in Code/Exercises/Exercise3.py

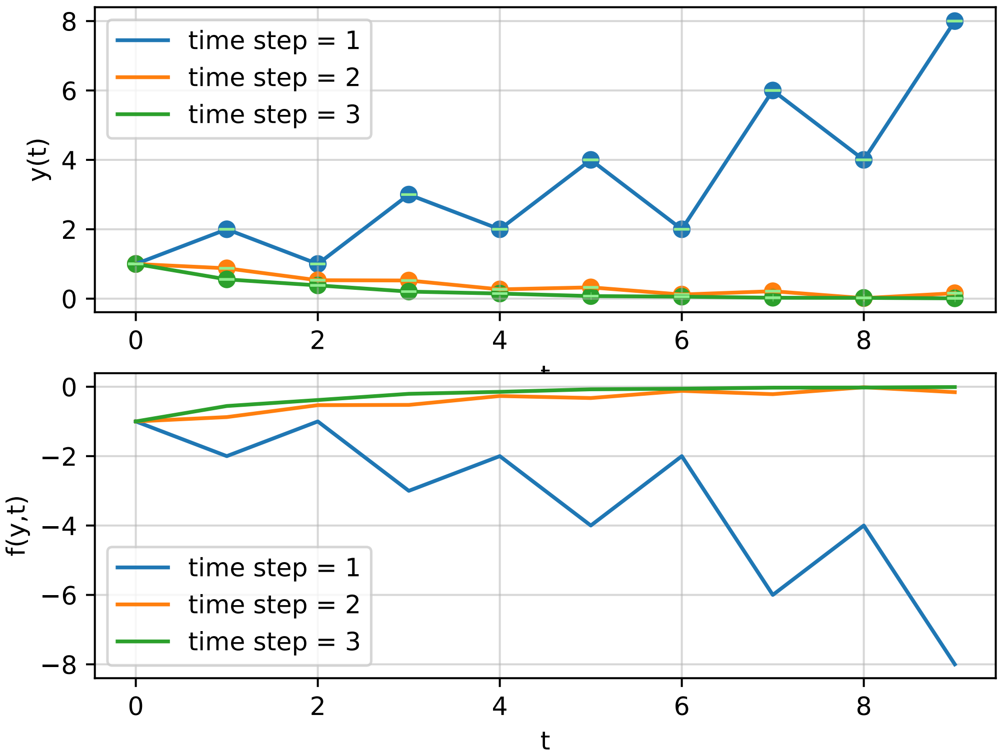

Figure 2: Figure showing the function $\frac{dy}{dt} = -y$, $y(0) = 1$, analysed numerically using the backward, or implicit Euler method and the Newton-Raphson method, with three different time steps. The upper pannel is the function and the lower pannel, its slope. The errorbars are shown in light green.

1.4 Exercise 4

2 Nuclear Reaction Networks

2.1 Exercise 5

Reaction 8 :

$$^1_1p + ^1_1p \rightarrow ^2_1d$$

It is missing two particles : a positron for the charge, as we have two positive charges on one size and only one on the other side. This however creates an imbalance in lepton number, and so we must add a particle without charge : an electron neutrino. The relation becomes:

$$^1_1p + ^1_1p \rightarrow ^2_1d + e^+ + \nu_e.$$

Reaction 9 :

$$^1_1p + ^7_3Li \rightarrow 2^4_2He$$

This equation is mostly balanced : there are 8 nucleons and 4 charges on both sides, but it is missing a lepton on the right hand side. The relation becomes :

$$^1_1p + ^7_3Li \rightarrow 2^4_2He + \nu_e$$

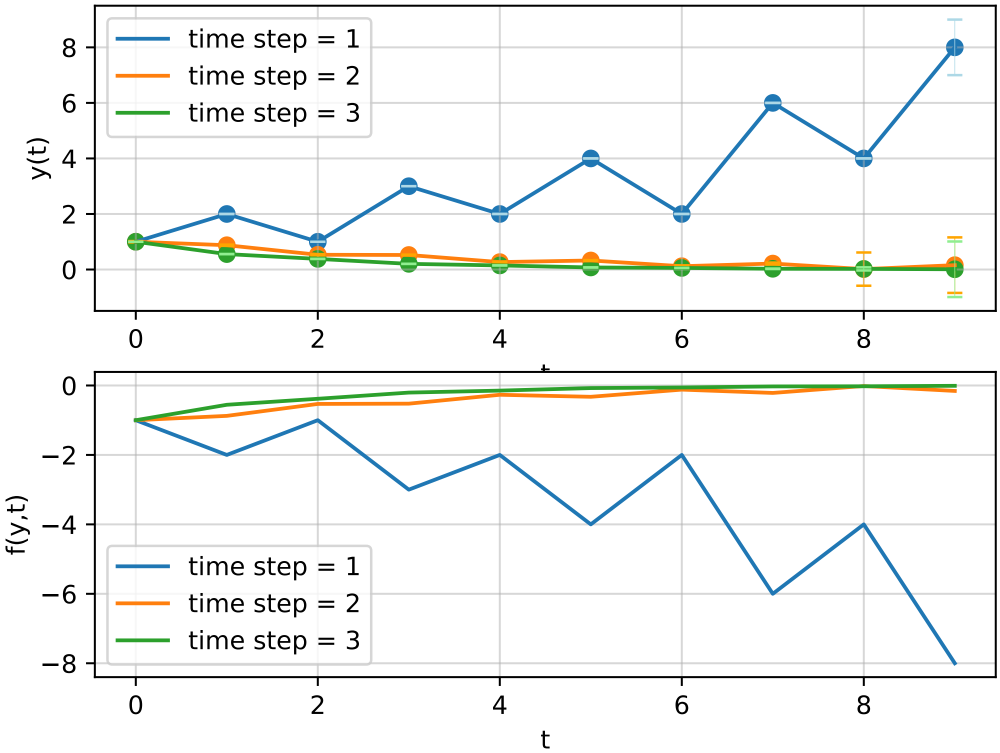

Figure 3: Figure showing the function $\frac{dy}{dt} = \frac{1}{(y^2+1)}$, $y(0) = 1$, analysed numerically using the backward, or implicit Euler method and the Newton-Raphson method, with three different time steps. The upper pannel is the function and the lower pannel, its slope. The errorbars shown in light green are the relative errors from one time step to the next.

Reaction 12 :

$$3{}^4_2He \rightarrow {}^{12}_6C$$

This reaction is already balanced : there are 12 nucleons, 6 charges and no leptons on both sides.

2.2 Exercise 6

$$\begin{aligned} \frac{dn_p}{dt} &= \\ \frac{dn_H}{dt} &= \\ \frac{dn_{He}}{dt} &= \end{aligned} \tag{1}$$

2.3 Exercise 7

a. There are 4 types of particles, and so 4 equations are needed to describe the system.
b.
c.